

Lecture 8 (10/2)

Correlation (2.4), Estimators, Bias, and Consistency

Standardized variables

Let X and Y be two random variables related by a linear regression model,
 $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$.

Standardized variables

Let X and Y be two random variables related by a linear regression model,
 $Y | X \sim \text{Normal}(\beta_0 + \beta_1 X, \sigma_\epsilon^2)$.

Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of Z_Y on Z_X ?

Standardized variables

$$Z_Y = (Y - \mu_Y) / \sigma_Y$$

Standardized variables

$$\begin{aligned} Z_Y &= (Y - \mu_Y) / \sigma_Y \\ &= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)] / \sigma_Y \quad (\text{substitute } Y, \mu_Y) \end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y \quad (\text{substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y \quad (\beta_0 \text{ cancels})\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1)\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X)\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y} && (Z_X = (X - \mu_X)/\sigma_X)\end{aligned}$$

Standardized variables

$$\begin{aligned}Z_Y &= (Y - \mu_Y)/\sigma_Y \\&= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)]/\sigma_Y && \text{(substitute } Y, \mu_Y) \\&= [\cancel{\beta_0} + \beta_1 X + \epsilon - \cancel{\beta_0} - \beta_1 \mu_X]/\sigma_Y && (\beta_0 \text{ cancels}) \\&= [\beta_1(X - \mu_X) + \epsilon]/\sigma_Y && \text{(factor out } \beta_1) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} \cdot \frac{X - \mu_X}{\sigma_X} + \frac{\epsilon}{\sigma_Y} && \text{(multiply, divide by } \sigma_X) \\&= \frac{\beta_1 \sigma_X}{\sigma_Y} Z_X + \frac{\epsilon}{\sigma_Y} && (Z_X = (X - \mu_X)/\sigma_X)\end{aligned}$$

Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted ρ .

Coefficient of determination

$$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 \quad (\text{variance of } Y = \beta_0 + \beta_1 X + \epsilon)$$

Coefficient of determination

$$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 \quad (\text{variance of } Y = \beta_0 + \beta_1 X + \epsilon)$$

$$1 = \frac{\beta_1^2 \sigma_X^2}{\sigma_Y^2} + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \quad (\text{divide by } \sigma_Y^2)$$

Coefficient of determination

$$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 \quad (\text{variance of } Y = \beta_0 + \beta_1 X + \epsilon)$$

$$1 = \frac{\beta_1^2 \sigma_X^2}{\sigma_Y^2} + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \quad (\text{divide by } \sigma_Y^2)$$

$$1 = \rho^2 + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \quad (\rho = \beta_1 \sigma_X / \sigma_Y)$$

Coefficient of determination

$$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 \quad (\text{variance of } Y = \beta_0 + \beta_1 X + \epsilon)$$

$$1 = \frac{\beta_1^2 \sigma_X^2}{\sigma_Y^2} + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \quad (\text{divide by } \sigma_Y^2)$$

$$1 = \rho^2 + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \quad (\rho = \beta_1 \sigma_X / \sigma_Y)$$

Therefore

$$1 - \rho^2 = \frac{\sigma_\epsilon^2}{\sigma_Y^2} = \frac{\text{var}(Y | X)}{\text{var}(Y)}$$

Coefficient of determination

$$\sigma_Y^2 = \beta_1^2 \sigma_X^2 + \sigma_\epsilon^2 \quad (\text{variance of } Y = \beta_0 + \beta_1 X + \epsilon)$$

$$1 = \frac{\beta_1^2 \sigma_X^2}{\sigma_Y^2} + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \quad (\text{divide by } \sigma_Y^2)$$

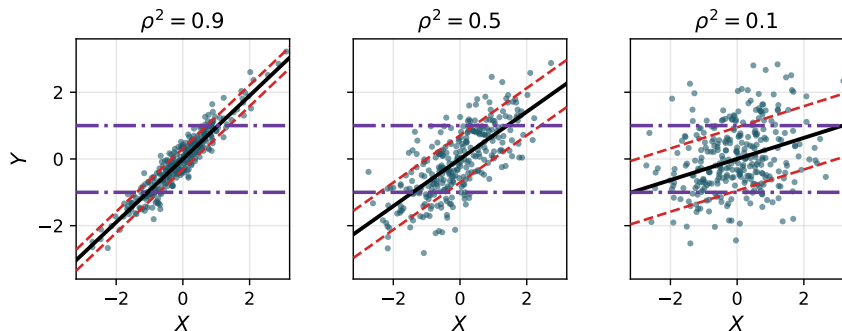
$$1 = \rho^2 + \frac{\sigma_\epsilon^2}{\sigma_Y^2} \quad (\rho = \beta_1 \sigma_X / \sigma_Y)$$

Therefore

$$1 - \rho^2 = \frac{\sigma_\epsilon^2}{\sigma_Y^2} = \frac{\text{var}(Y | X)}{\text{var}(Y)}$$

ρ^2 is called the coefficient of determination

Visualizing ρ^2



Here $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$ (hence $\text{cov}(X, Y) = \beta_1 = \rho$). The black lines show $E[Y | X] = \rho X$ and red lines $\pm \sigma_\epsilon = \pm \sqrt{1 - \rho^2}$. Purple: $\pm \sigma_Y = \pm 1$.

Summary: measures of association

	Formula	Units	Range
$\text{cov}(X, Y)$		units $Y \times$ units X	$(-\infty, \infty)$
β_1	$\frac{\text{cov}(X, Y)}{\sigma_X^2}$	units $Y/\text{units } X$	$(-\infty, \infty)$
ρ	$\frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$	none	$[-1, 1]$
ρ^2	$1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2}$	none	$[0, 1]$

Estimators

An estimator $\hat{\theta}$ of a parameter θ is any function of the data meant to approximate θ .

Estimators

An estimator $\hat{\theta}$ of a parameter θ is any function of the data meant to approximate θ .

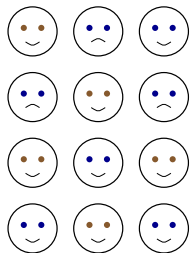
- $\hat{\theta}$ is random – it will vary depending on which sample we collect.

Estimators

An estimator $\hat{\theta}$ of a parameter θ is any function of the data meant to approximate θ .

- $\hat{\theta}$ is random – it will vary depending on which sample we collect.
- θ is a fixed, non-random number (in frequentist statistics; we will change this in Unit 6).

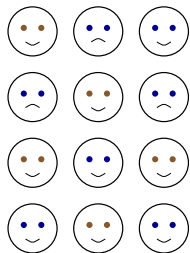
Replicates



Replicate 1

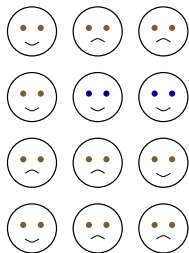
$$\hat{q} = 3/12$$

Replicates



Replicate 1

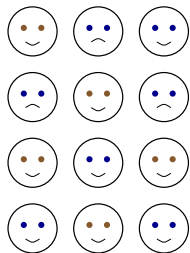
$$\hat{q} = 3/12$$



Replicate 2

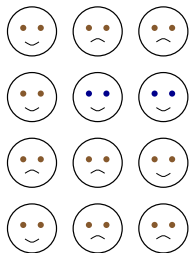
$$\hat{q} = 6/12$$

Replicates



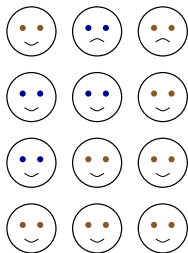
Replicate 1

$$\hat{q} = 3/12$$



Replicate 2

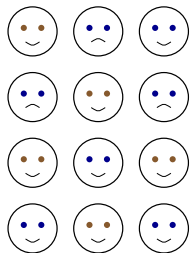
$$\hat{q} = 6/12$$



Replicate 3

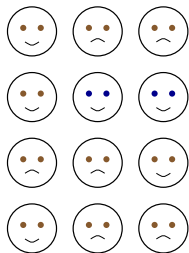
$$\hat{q} = 2/12$$

Replicates



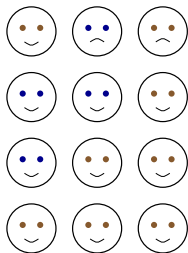
Replicate 1

$$\hat{q} = 3/12$$



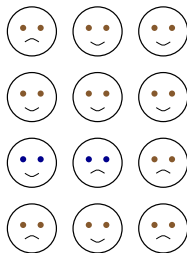
Replicate 2

$$\hat{q} = 6/12$$



Replicate 3

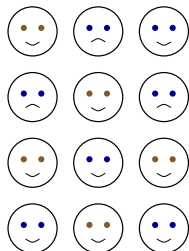
$$\hat{q} = 2/12$$



Replicate 4

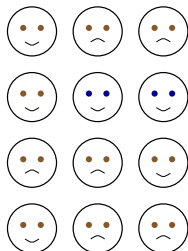
$$\hat{q} = 5/12$$

Replicates



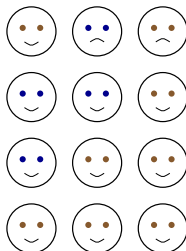
Replicate 1

$$\hat{q} = 3/12$$



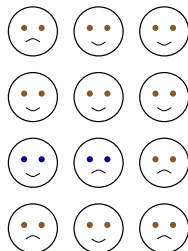
Replicate 2

$$\hat{q} = 6/12$$



Replicate 3

$$\hat{q} = 2/12$$



Replicate 4

$$\hat{q} = 5/12$$

$\hat{q}$ changes from replicate to replicate, while q stays fixed. The distribution of $\hat{q}$ over (infinitely many) replicates is its sample distribution; its spread is $se(\hat{q})$.

Sample distribution and standard error

Example

Let $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ and estimate μ with $\hat{\mu} = \bar{Y} = \frac{1}{N} \sum_i Y_i$.

Sample distribution and standard error

Example

Let $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ and estimate μ with $\hat{\mu} = \bar{Y} = \frac{1}{N} \sum_i Y_i$.

A sum of Normals is Normal, so we know the distribution of $\hat{\mu}$ exactly:

$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

Sample distribution and standard error

Example

Let $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ and estimate μ with $\hat{\mu} = \bar{Y} = \frac{1}{N} \sum_i Y_i$.

A sum of Normals is Normal, so we know the distribution of $\hat{\mu}$ exactly:

$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

The sample distribution is the distribution of $\hat{\mu}$ over replicates of the data. The standard deviation of the sample distribution is the standard error

$$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

Sample distribution and standard error

Example

Let $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ and estimate μ with $\hat{\mu} = \bar{Y} = \frac{1}{N} \sum_i Y_i$.

A sum of Normals is Normal, so we know the distribution of $\hat{\mu}$ exactly:

$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

The sample distribution is the distribution of $\hat{\mu}$ over replicates of the data. The standard deviation of the sample distribution is the standard error

$$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

Note: We need to remember that in reality we don't know σ , so this itself needs to be estimated. Some people call the distribution with estimated σ the sample distribution.

Estimating q

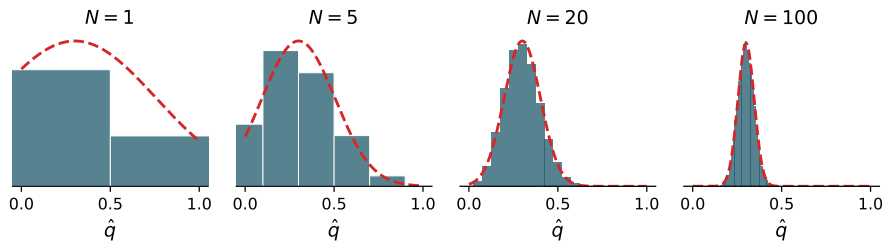
Example

Let $Y_1, \dots, Y_N \sim \text{Bernoulli}(q)$ and estimate q with $\hat{q} = \overline{Y}$.

Estimating q

Example

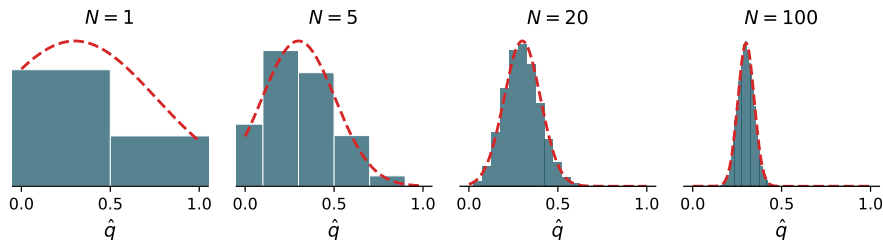
Let $Y_1, \dots, Y_N \sim \text{Bernoulli}(q)$ and estimate q with $\hat{q} = \overline{Y}$.



Estimating q

Example

Let $Y_1, \dots, Y_N \sim \text{Bernoulli}(q)$ and estimate q with $\hat{q} = \bar{Y}$.



$\hat{q}$ is not Normal, but by the CLT the Normal result carries over for large N , with $\sigma^2 = q(1 - q)$:

$$\hat{q} \approx \text{Normal} \left(q, \frac{q(1 - q)}{N} \right), \quad \text{se}(\hat{q}) = \sqrt{\frac{q(1 - q)}{N}}.$$